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Wealth of Retailers: Implications for Sourcing
Strategies and Supply Chain Governance

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Abstract

Supply chain and reputational risks are often assumed to motivate firms to source production in developed, high-cost countries rather than developing, low-cost countries. To examine this assumption, we provide evidence from the collapse of the Rana Plaza building on April 24, 2013, which with its 1,133 fatalities and 2,438 injuries is seen as one of the worst industrial accidents in history. This paper analyzes the stock market reaction to the Rana Plaza disaster in the Bangladeshi ready-made garment industry. Sufficiently negative stock market reaction might motivate firms to shift their sourcing strategy. Our analysis is based on a sample of 39 publicly-traded global apparel retailers with significant garment sourcing in Bangladesh. Although stock market reaction to retailers on the day of the Rana Plaza disaster is negative, its magnitude and significance dissipates when considering the following day's stock market reaction. There is no evidence of significant stock market reaction over 11 trading days (approximately two weeks in calendar time) after the Rana Plaza disaster. Subsequent to the disaster, retailers developed two different agreements to improve factory and worker safety in Bangladesh – the Accord on Fire and Building Safety in Bangladesh (AFBSB), and the Alliance for Bangladesh Worker Safety (ABWS). We find no evidence of significant stock market reaction to the announcements of the AFBSB and the ABWS. From a retailer perspective, since negative economic impacts from the Rana Plaza disaster were insignificant, they have little economic incentive to move sourcing out of Bangladesh or other low-cost countries. We discuss the implications of our results for retailers, non-governmental organizations (NGOs), garment factory owners in Bangladesh, the Bangladeshi government, as well as academic researchers.

Keywords: *responsible sourcing, supply chain governance, corporate reputation, stock market reaction, industrial accidents*

1. Introduction

At approximately 9:00 AM local time on Wednesday, April 24, 2013, Rana Plaza, an eight-story commercial building in Savar, Bangladesh, collapsed. The building housed several garment factories employing approximately 5,000 workers. Due to the nature of the disaster site and the caution required to not further injure survivors, recovery efforts took several days; the final survivor was not rescued until May 10, 16 days after the building collapse. Despite immediate local and international rescue response, 1,133 fatalities and 2,438 injuries were suffered (CPD 2013). Although garment factories in Bangladesh and other developing countries had previously experienced industrial accidents, the Rana Plaza disaster is one of the worst industrial accidents in history (Hobson 2013). The scale of this unspeakable tragedy increased awareness of the risks and costs of sourcing from low-cost countries, as well as the need to improve supply chain governance mechanisms for working conditions and safety in the global garment industry.

A commonly held assumption is that, for firms sourcing in developing countries like Bangladesh, incidents like the Rana Plaza disaster cause enough financial and reputational damage to motivate firms to either: 1) move production to more developed, high-cost countries; or 2) actively engage in improving working conditions in the developing, low-cost country. Further, the media and risk exposure resulting from the Rana Plaza disaster suggests that protection from such exposure might represent an advantage for producers in developed economies where worker safety is well managed. In this paper, we study whether such an advantage exists by estimating the shareholder wealth effects of the Rana Plaza disaster for firms with significant apparel sourcing contracts in Bangladesh. The magnitude of economic impact can provide insights into retailer willingness-to-pay a sufficient premium to ensure safe working conditions and/or to justify a change in sourcing strategy.

Our research approach builds upon the work of Hendricks and Singhal (2003, 2005a, 2005b) who find the shareholder wealth impacts of supply chain disruptions to be significantly negative and typically greater in magnitude than many other types of operational, marketing, and financial events (Hendricks and Singhal 2003, Table 4). Their results provide evidence that investors are concerned about supply chain risks, and react when risks are realized. For the supply chain events studied by Hendricks and Singhal, investors can determine responsibility of the focal firms and/or their supply chain partners. However, in the case of Rana Plaza, the responsibility of retailers is ambiguous. In this paper, we document the economic impacts of the

Rana Plaza disaster, and thereby inform policy and practice of the effectiveness of market forces in motivating the improvement of working conditions in developing countries and avoiding future tragedies. Better understanding of the economic consequences could also inform other stakeholders (e.g., factory owners, governments, NGOs) about the role of the financial markets in affecting change in low-cost country working conditions, and potentially what their own roles and responsibilities should be in developing better supply chain governance strategies.

Improvements to working conditions require engagement and collaboration among stakeholders including national governments, factory owners, buyers, and non-governmental organizations (NGOs). It is unclear what mechanisms will be effective in mitigating industrial accidents and monitoring safety in the garment industry, and how the responsibility for worker safety will be distributed among the stakeholders. In the aftermath of the Rana Plaza disaster, retailers took the brunt of the criticisms for the safety conditions in the Bangladeshi garment industry (e.g., Blair 2013, Silcoff 2013). Labor unions pointed fingers at retailers for ignoring safety violations in factories, and – together with NGOs – argued that retailers should take responsibility for working conditions in Bangladeshi factories. The failure of retailers to take action was claimed by labor unions to amount to criminal negligence (CCC 2013). The popular press and social media also criticized retailers for their inability to ensure safe working conditions, contending that retailers have a moral and ethical obligation to offer such conditions. Others countered that retailers were being made the scapegoats, and that blame should instead belong to the indifferent factory owners who neglected safety, and the national and local governments that failed to enforce industrial and labor laws (Bhagwati and Narlikar 2013). Buyers, as key stakeholders, thus must consider the risk that worker safety entails, especially given that no one yet has a clear picture of how to ensure safe working conditions. Accordingly, we also study the shareholder wealth effects of the retailer responses to the Rana Plaza disaster. Although major retailers initially discussed a single coordinated industry response, two different industry agreements emerged. The Accord on Fire and Building Safety in Bangladesh (hereafter referred to as AFBSB) was established by a group dominated by European retailers, international labor organizations, and NGOs. An alternative agreement, the Alliance for Bangladesh Worker Safety (hereafter referred to as ABWS) was established by North American retailers.

The Rana Plaza disaster is important to study not just because of the human scale of the tragedy but also due to the prominence of Bangladeshi exports in the global garment industry. Global apparel retailers (hereafter referred to as *retailers*) are skilled at finding and sourcing garments from low-cost production locations. In the past, China was the primary location for garment sourcing. As China started experiencing wage increases and capacity pressures, retailers looked for alternative production sources. Bangladesh emerged as an attractive location with its abundant supply of labor, with labor costs among the lowest in the world. As an added bonus, governments in many developed economies lowered their import tariffs from Bangladesh to spur its economic development, further reducing the cost of sourcing from Bangladesh. A 2011 McKinsey & Company survey of Chief Purchasing Officers at European and US apparel companies indicated that 86% would decrease or strongly decrease sourcing from China within the next five years, and that Bangladesh was their preferred alternate location (Berg et al. 2011). Consequently, the Bangladeshi garment industry has become a major exporter to the West, second only to China. It employs approximately 4 million Bangladeshis in over 5,000 factories, and accounts for 45% of the country's total industrial employment. In 2012, the Bangladeshi garment industry exported about \$20 billion of garments, which accounted for nearly 75% of national export earnings and 10% of GDP (BGMEA 2014). Approximately 75% of the exports were to European retailers and 25% to North American retailers (OEC 2014). Hennes & Mauritz in Europe and Wal-Mart in North America are the largest importers of garments from Bangladesh (D'Innocenzio 2013).

The results of our study are based on an analysis of the stock market effects on 39 publicly-traded retailers that signed either the AFBSB or ABWS within 60 trading days (approximately three months in calendar time) following the Rana Plaza disaster. Although stock market reaction to retailers on the day of the Rana Plaza disaster is negative and significant, its magnitude and significance dissipates when considering the following day's stock market reaction. There is no evidence of significant stock market reaction over 11 trading days (approximately two weeks in calendar time) after the Rana Plaza disaster. We find no evidence of a significant stock market reaction to the announcements of the AFBSB and the ABWS. This could indicate that investors view these agreements as not that different from what retailers were already doing to improve working conditions and safety in Bangladeshi garment factories. Further, over 60 trading days after the Rana Plaza disaster, the stock market reaction adjusted for other

contemporaneous announcements unrelated to the Rana Plaza disaster is statistically insignificant. These results suggest that investors do not see the Rana Plaza disaster as seriously damaging the reputation of retailers that source from Bangladesh and/or significantly disrupting their apparel supply chains.

The lack of significant stock market reactions may be surprising to many, especially in light of the significant negative impacts of other supply chain disruptions (Hendricks and Singhal 2003, 2005a, 2005b). From a retailer perspective, since negative economic impacts from the Rana Plaza disaster were insignificant, they have little economic incentive to move sourcing out of Bangladesh. Further, our findings have important policy implications for governments, NGOs, consumers, and other stakeholders seeking to understand whether incidents such as the Rana Plaza disaster shift the competitive playing field away from low-cost countries, and also whether market forces alone are sufficient to improve working conditions in developing countries and avoid future tragedies.

Section 2 discusses how disasters like Rana Plaza can potentially impact the shareholder wealth of buying firms. Section 3 describes the sample and methodology. Section 4 presents the results. Section 5 summarizes and discusses the similarities and differences of our findings from existing research, and the implications of our findings for various stakeholders.

2. Shareholder Wealth Impacts of Rana Plaza

There are at least two major mechanisms for disasters like Rana Plaza to potentially impact the shareholder wealth of buying firms: damage to reputation; and supply disruptions.

2.1 Reputation

Corporate reputation is important as it can signal to consumers and investors such key attributes as product quality, employment desirability, investment quality, strategy efficacy, and corporate social responsibility (Roberts and Dowling 2002). Eberl and Schwaiger (2005) argue that reputation is an intangible asset that is difficult to imitate, and thus a source of competitive advantage. Roberts and Dowling (2002) and Eberl and Schwaiger (2005), among others, find that reputation is positively linked to financial performance.

Corporate reputations are difficult to build, requiring persistent time, effort, and management resources (Fombrun and Shanley 1990). Unfortunately, reputations are easily damaged and difficult to repair (Rhee and Valdez 2009). Association with accidents and disasters can damage corporate reputations, particularly

when a firm is perceived to be responsible for the damaging event (Coombs and Holladay 2002). For example, Maloney and Mulherin (2003) discuss the potential reputational damage to the four supplier firms associated with the space shuttle *Challenger* disaster; they find that although all four firms suffered initially negative stock market reactions, the firm with primary responsibility experienced significantly greater negative stock returns.

Another major determinant of corporate reputation is social performance. Turban and Greening (1997) posit that good social performance enhances the firm's reputation. Brammer and Pavelin (2006) find that good social performance improves corporate reputation, and that the link is stronger in industries with salient social issues. As McWilliams and Siegel (2001, p. 120) state: "...support of corporate social responsibility creates a reputation that a firm is reliable and honest. Consumers typically assume that the products of a reliable and honest firm will be of high quality." Several researchers examine the links between corporate social performance, reputation, and financial performance. In a meta-analysis of 52 studies, Orlitzky et al. (2003) find that corporate social performance and financial performance are positively related, and conclude that a large portion of social performance benefits are due to its effect on improving reputation. Researchers generally argue and demonstrate that the stock market reacts positively to instances of good social performance, and negatively to instances of poor social performance. For example, Klassen and McLaughlin (1996) demonstrate a positive stock market reaction to environmental awards and a negative reaction to environmental crises. Jacobs et al. (2010) find a positive stock market reaction to environmental philanthropy and ISO 14000 certifications. The effects of poor social performance are demonstrated by Davidson and Worrell (1988) in their study of corporate illegalities.

Given the history of adverse publicity from social issues in sourcing such as those experienced by Nike and Levi-Strauss in the 1990s, many retailers developed and implemented sourcing policies and codes of conduct for suppliers that emphasize social and environmental issues (Doorey 2011). However, disasters such as Rana Plaza can undermine retailer claims about socially responsible sourcing. The Rana Plaza disaster sparked outrage about working conditions and safety in Bangladeshi garment factories and the lack of oversight by apparel retailers of their Bangladeshi suppliers. Given the extensive media coverage of the disaster, retailers faced a high level of corporate reputational risk by being associated with dangerous and unsafe working conditions in Bangladesh. Scott Nova of the Worker Rights Consortium reacted to the

Rana Plaza disaster by stating “The apparel brands and retailers face a greater level of reputation risk of being associated with abusive and dangerous conditions in Bangladesh than ever before” (Greenhouse 2013, pg. 1).

Given that poor reputation can hurt financial performance, and that association with disasters like Rana Plaza contributes to poor reputation, we expect that buying firms will be negatively impacted by the Rana Plaza disaster.

2.2 Supply disruptions

In a retail environment, the direct effects of supply disruptions can be stock-outs, lost sales, and reduced market share. However, the limited number of Bangladeshi factories located at Rana Plaza together with the available capacity of alternate sources could limit negative direct effects. More important are the indirect effects of supply disruptions that can include increased costs for expediting and other recovery measures such as the development of alternate sourcing. Supply disruptions can also prompt firms to alter their sourcing strategies, including number of tiers, supplier development, and geographic locations of suppliers, all of which can affect the effectiveness and cost of sourcing. For example, despite the labor cost advantages, Nike decided to move most of its sourcing out of Bangladesh, and Disney ceased all its Bangladeshi manufacturing (Banjo 2014).

Accidents such as the Rana Plaza disaster can impose other supply chain related costs. For example, governments can increase regulations to avoid future accidents, which can increase manufacturing costs. As happened in Bangladesh (INHT 2013), anger and frustration that follow such industrial accidents can lead to protests, strikes, and work stoppages which can further disrupt supply chains. Retailers may also have to absorb the cost of fixing safety issues and poor working conditions, as well as the increased supply chain governance and oversight required to avoid future accidents.

In the Rana Plaza case, some retailers were surprised that their garments were being made in factories at Rana Plaza even though these factories were not authorized suppliers (Labowitz and Baumann-Pauly 2014). In such instances, the primary supplier had subcontracted the production without the knowledge of the retailer. Unauthorized subcontracting raises the issue of how much control and visibility retailers have over their supply chains, including their ability to track orders and production. This could create

uncertainties in the mind of customers and investors about the overall capabilities and effectiveness of retailers' supply chains, and could affect the retailers' future sales and valuation.

The financial effects of supply disruptions are well-known. Hendricks and Singhal (2003) estimate that supply disruptions result in an average -10.3% stock market reaction. They find that the magnitude of the disruption reaction is similar for all types of industries including retail, and is independent of the disruption source, whether due to internal, customer, supplier, or other reasons. Although it can be argued that the risks of sourcing in Bangladesh were well-known prior to the Rana Plaza disaster, and the stock markets might have anticipated incidents such as Rana Plaza based on previous accidents and knowledge about Bangladeshi working conditions, markets cannot anticipate when accidents will occur, where they will occur, nor the scope and severity of such accidents. Risks are inherent in many supply chain activities whether it is sourcing in Bangladesh, operating in flood-prone regions, or investing in unproven technologies. However, it is not until such risks are actually realized, that markets typically react.

To summarize, given that the Rana Plaza disaster can negatively affect reputation and disrupt supply chains, our first hypothesis is:

H1. The stock market reaction to the Rana Plaza disaster is negative for firms that source garments in Bangladesh.

2.3 Retailers' response strategy

Prior to the Rana Plaza disaster, many major retailers established supplier codes of conduct and monitored the compliance of their suppliers in Bangladesh and other developing nations. In some cases, these efforts were independent, and in others, they were conducted in collaboration with trade groups. For example, Wal-Mart independently established its ethical sourcing strategy in 1992 to hold suppliers accountable for safety standards and practices of their subcontractors. In 2012, Wal-Mart conducted audits at over 10,000 suppliers as part of this program (Wal-Mart 2013). As an example of collaborative efforts, PVH, Tchibo, and others signed the Bangladesh Fire and Building Safety Agreement in 2012. The agreement mandated inspections of supplier factories, public reporting of inspection reports, mandatory repairs to fix hazards, and a threat that non-compliant suppliers would be cut off (ILRF 2012). Although well-intentioned, the existence of multiple codes and programs presents challenges of coordination and standardization for both the retailers and suppliers (Locke 2013).

The Rana Plaza disaster prompted retailers to discuss coordinated response strategies to prevent recurrences. Two multi-firm agreements were negotiated. A group dominated by European retailers, international labor organizations, and NGOs developed the Accord on Fire and Building Safety in Bangladesh (AFBSB), a five-year accord that establishes an inspection system for Bangladeshi factories, and contains several options for member firms to financially assist in factory safety improvements. This accord also contains an arbitrated dispute resolution process. It was announced on May 11, 2013, just over two weeks after the Rana Plaza disaster.

Citing concerns with the stance of the AFBSB on contractual issues such as sourcing and prices, its dispute resolution process, and the different legal liability standards in the US, many North American firms declined to sign the AFBSB and instead crafted their own agreement. The result is the Alliance for Bangladesh Worker Safety (ABWS); it is an agreement that establishes a \$42M fund to be committed by the signees over a five-year period, as well as a \$100M low-interest loan pool for Bangladeshi factory owners to access for needed safety improvements. It was announced on July 8, 2013, 10 weeks after the disaster. The ABWS received criticism from labor groups for its lack of stringency and non-binding structure (Greenhouse and Clifford 2013).

Table 1 summarizes the major features of each agreement. While much has been made of the differences between the two, both agreements establish training, inspection, and remediation programs to improve building and fire safety in Bangladeshi garment factories. They also both advocate transparency in identification and monitoring of supplier factories. In fact, a report by the NYU Center for Business and Human Rights concludes that the two agreements are “fundamentally similar” (Labowitz and Baumann-Pauly 2014, p. 10).

Compared with pre-disaster efforts, both agreements are better funded and coordinated, more far-reaching, and more proactive. Coupled with increased scrutiny from NGOs and the public, the likelihood of program success is improved. As a result, this can reduce the potential for industrial accidents. By sharing information about suppliers and their compliance, the agreements increase supply chain visibility and transparency for both the retailers and suppliers. Consolidation and standardization of inspection and compliance efforts can also reduce supplier confusion. Taken together, these factors enable improved supply chain coordination and governance. Further, although the agreements require financial contributions

from the member firms, the pooling of resources can reduce costs that result from duplication of retailer efforts in establishing individual codes of conduct, inspection programs, and remediation efforts. The expected improvements in supply chain governance as well as reductions in total monitoring costs can result in a positive stock market reaction. Accordingly, our hypothesis is:

H2. The stock market reactions to announcements of the AFBSB and ABWS agreements are positive for AFBSB and ABWS signees.

3. Sample and Methodology

3.1 Sample

In developing the sample, our premise is that the Rana Plaza disaster puts the spotlight on all retailers sourcing from Bangladesh, not just those dealing directly with Rana Plaza. In the aftermath of the Rana Plaza incident, many reports and press articles highlighted the major challenges that retailers face in sourcing from Bangladesh (e.g., Labowitz and Baumann-Pauly 2014, Srivastava and Shannon 2013, Cendrowski 2013). These reports argue that the poor workplace conditions and factory safety violations at Rana Plaza were not unique to that facility but instead are widespread throughout the Bangladeshi apparel industry. Further, the articles highlight unauthorized subcontracting as one of the major issues with Bangladeshi suppliers. Given the inability to prevent such subcontracting, sourcing within Bangladesh potentially exposes buyers to factories like Rana Plaza, whether or not they directly source at Rana Plaza. Thus, we expect that even firms without direct Rana Plaza suppliers will be affected by the Rana Plaza incident as investors may associate similar problems with other Bangladeshi sources. While the size and number of sourcing contracts to specific suppliers is not publicly available, there are other ways that investors and other stakeholders can learn about a retailer's sourcing relationships in Bangladesh. These include analyst reports, business press articles, corporate press releases and websites, import/export records, and country-of-origin labeling on garments.

Our sample is based on retailers that signed either the AFBSB or the ABWS. To justify the appropriateness of our sample, Figure 1 presents a four quadrant matrix that segments retailers based on whether they signed AFBSB or the ABWS or not, and whether they source from Bangladesh or not. Although all firms that source in Bangladesh are likely affected by the Rana Plaza disaster, firms that signed either the AFBSB or the ABWS (Figure 1 quadrant 1) are probably the most impacted. Not only are these firms subject to possible supply disruptions but their public participation in an agreement conveys some

level of responsibility and heightens the potential for reputational impact. While it is possible that some firms buying garments from Bangladesh did not sign either agreement (Figure 1 quadrant 2), they probably depend less on exports from Bangladesh, and hence, any impact from the Rana Plaza disaster is expected to be less than that to firms in quadrant 1. Further, identifying all firms in quadrant 2 is problematic given the lack of a definitive database of sourcing information.

Firms not sourcing in Bangladesh are less affected by the Rana Plaza disaster than those who do. Although it is possible that firms not sourcing in Bangladesh did sign either the AFBSB or ABWS (Figure 1 quadrant 3), it seems unlikely given that both the AFBSB and ABWS bind the signees in various ways including commitment of resources, legal liability, etc. Firms that do not source in Bangladesh and did not sign an agreement (Figure 1 quadrant 4) are likely the least affected by the Rana Plaza disaster.

In accordance with the above discussion, our sample comprises firms in Figure 1 quadrant 1. As evidence that firms signing either the AFBSB or the ABWS have substantial sourcing activities in Bangladesh, each of the firms mentioned as customers of garment suppliers located in Rana Plaza during the days immediately following the disaster (e.g., Loblaw, Primark, Wal-Mart) are indeed initial signees to either the AFBSB or the ABWS. Further, examination of US import records in the Panjiva database confirms that each of the US signees received multiple shipments from Bangladeshi suppliers in 2013. Thus, our sample comprises all publicly-traded firms that signed either the AFBSB or the ABWS within 60 days of April 24, 2013. Although the majority of the AFBSB signees are European, some US retailers including American Eagle Outfitters and PVH, and Asian retailers including Esprit also signed the accord. All the ABWS signees are Canadian or US firms.

Table 2 lists the 25 publicly-traded firms that signed the AFBSB, and the 14 publicly-traded firms that signed the ABWS within 60 days of April 24, 2013. Table 3 presents descriptive statistics for the combined sample of 39 firms. US firms account for 38.5% of the combined sample and UK firms account for 17.9%. The two most commonly represented industries are general merchandise retailers (SIC 53) such as J.C. Penney and Marks & Spencer that make up 33.3% of the sample, and apparel retailers (SIC 56) such as Abercrombie & Fitch and Hennes & Mauritz (H&M) that represent 28.2% of the sample. Panel C of Table 3 indicates that the sample consists of large firms with mean (median) sales of \$40.9 billion (\$11.4 billion) in 2012. If we consider only North American firms, the 12 retailers that signed an agreement accounted

for 69.2% of all general and apparel retail sales in 2012 and include four of the five largest retailers by sales (Wal-Mart, Target, Sears, and Macy's).

3.2 Methodology

We use event study methodology to estimate the shareholder wealth effects of the Rana Plaza disaster. This methodology estimates “abnormal” stock market returns associated with specific events, while controlling for market-wide influences on stock prices (Brown and Warner 1985). The abnormal returns are an estimate of the change in stock price associated with an event. The basis of event study methodology is that, in an efficient market, the shareholder wealth effects of an event are immediately reflected in the stock price.

We estimate abnormal returns for five different time periods. The first period measures the immediate effects of the Rana Plaza disaster. The Rana Plaza building collapsed at 9:00 AM Bangladesh Standard Time on Wednesday, April 24, 2013 (11:00 PM Eastern Standard Time on April 23 and 3:00 AM Greenwich Mean Time on April 24). Thus, April 24, 2013 is the first trading day on the North American, European, and Asian markets that the stock market could react to the event. It is quite common in stock market reaction studies to estimate reaction on the day of the announcement and the day before or after and attribute this to the event under study (Fisher-Vanden and Thorburn 2011, Jacobs and Singhal 2014). However, in our case, information about the severity of the disaster gradually unfolded over a number of days. For example, the death tolls initially reported were less than 100 but gradually increased to over 1,100 as rescue efforts continued over the following weeks. The early days of the recovery also continued to bring news of additional firms that purchased from Rana Plaza or other Bangladeshi suppliers. Analysis of the business press indicates that press coverage peaked on April 26, 2013, and substantially declined by May 2, 2013 but the event continued to be covered in the press after May 2, 2013. To account for this ongoing information release, we estimate the stock market reaction on the day of the disaster as well as the subsequent two weeks. Given that a calendar week comprises five trading days, our event period comprises 11 trading days from April 24, 2013 until May 8, 2013.

We translate calendar days into event days such that Day 0 is the day of the event, Day 1 is the trading day following the event, Day -1 is the trading day before the event, etc. For the first period, April 24, 2013 is Day 0 and May 8, 2013 is Day 10. Event days represent the number of trading days relative to a given

event, not necessarily the same calendar dates. This is important to note as our sample comprises firms in several stock markets, and not every stock market is open on all the same calendar dates due to differing national holidays and other considerations. For convenience, we report US calendar dates that the market is open in our discussions and results.

The second period measures the effect of the announcement of the AFBSB. The first news regarding the specifics of the AFBSB was released in Bangladesh on Saturday, May 11, 2013 (Financial Express 2013). The first subsequent trading day is Monday, May 13, 2013, and is thus designated as Day 0 in our examination of the stock market reaction to the AFBSB release. Following the announcement of the AFBSB, analysts and experts discussed and debated the strengths, weaknesses, and implications of the AFBSB in the press. The additional information revealed by this ongoing media discussion could affect the stock market response in the days subsequent to the AFBSB announcement. Business press coverage peaked on May 14, 2013, and substantially declined by May 18, 2013. To capture these potential effects, we estimate the stock market reactions over a period of six trading days from May 13, 2013 until May 20, 2013 (Day 5).

The third period measures the effect of the announcement of the ABWS. The first news regarding the specifics of the ABWS was released on Monday, July 8, 2013 (McVeigh 2013) and so July 8 is designated as Day 0. As with the AFBSB announcement, there was discussion in the press about the ABWS and how it compares with the AFBSB. Analysis of the business press indicates that press coverage peaked on July 9, 2013, and substantially declined by July 13, 2013. To capture the potential effects of this discussion and debate about the ABWS, we estimate stock market reactions over a period of six trading days from July 8, 2013 until July 15, 2013 (Day 5).

We use an 11-day period to examine the effects of the disaster, and six-day periods to examine the effects of AFBSB and ABWS announcements due to the longer period over which additional information about the severity of the disaster unfolded. Hence, we use different period lengths to estimate the effects of the disaster versus the AFBSB and ABWS announcements. For robustness, we also estimated the effects of the AFBSB and ABWS announcements using 11-day periods and the conclusions are the same.

The fourth period measures the cumulative effects of our first three periods. We estimate abnormal returns over 23 days by aggregating the 11-day first period and the six-day second and third periods. Thus,

the fourth period combines the total stock market effects of the Rana Plaza disaster as well as the announcements of the two agreements.

For completeness, we also include a fifth period that measures the abnormal returns over the 60 consecutive trading days beginning on April 24, 2013 (Day 0) and ending on July 19, 2013 (Day 60). This fifth period comprises the 23 days included in the fourth period, and also considers the effect of any information that may have been released outside of the 23-day periods. Figure 2 illustrates the timeline of the five event periods that we consider.

Consistent with many event studies (Fisher-Vanden and Thorburn 2011, Jacobs and Singhal 2014), we use the “market model” to estimate abnormal returns. Appendix A gives the methodology details. Since our sample of 39 firms is somewhat small, mean results can be unduly influenced by outliers. Accordingly, we emphasize median and percent negative (positive) abnormal returns in our discussion. We test for the statistical significance of the median abnormal return using the Wilcoxon signed-rank test. We use the binomial sign test to determine if the percent negative (positive) abnormal returns are significantly different than the null of 50%. For all tests, we report two-tailed *p*-values.

Given the length of event periods (6, 11, 23, and 60 days) over which we are estimating abnormal returns, a natural question is the extent to which estimated abnormal returns might be due to other contemporaneous announcements made by our sample firms that are unrelated to the Rana Plaza disaster. To address this issue, we adjust the time series of each sample firm’s abnormal returns for other announcements that are known to potentially affect shareholder wealth. Appendix B provides the details of this adjustment. We report results for both the unadjusted and the adjusted samples.

4. Results

4.1 Stock market reaction to the Rana Plaza disaster

Table 4 Panel A presents the unadjusted abnormal returns on the day of the Rana Plaza disaster (Day 0) and each of the ten succeeding days for the combined sample of AFBSB and ABWS signees. As mentioned earlier, we emphasize median and percent negative (positive) abnormal returns in our discussion rather than the mean, because mean results can be unduly influenced by outliers in our somewhat small sample. The Day 0 median abnormal return is -0.40% , significantly different from zero at the 5% level; 26 out of the 39 firms (66.67% of sample firms) experience negative abnormal returns on Day 0,

significantly different from 50% at the 5% level; but the mean abnormal return is insignificant. The mean, median, and percent negative abnormal returns for the day after the disaster (Day 1) are slightly positive but insignificant. Considering the two-day period of Day 0 and Day 1, the mean, median, and percent negative cumulative abnormal returns (CARs) are insignificant. The abnormal returns for the remaining nine days (Day 2 through Day 10) are all insignificant. Table 4 Panel A also presents the CARs for Days (0, 5), the event day and the first trading week; Days (6, 10), the second trading week; and Days (0, 10), the entire 11-day period. None of the CARs are statistically significant.

Panel B of Table 4 presents the adjusted abnormal returns. We identify 36 contemporaneous announcements for the 39 sample firms over the 11-day period that are unrelated to the Rana Plaza disaster, and adjust the abnormal returns by removing the abnormal returns around these announcements. After adjusting for the potential confounding effects of other contemporaneous announcements, the results are substantively similar to those presented in Panel A. The Day 0 abnormal returns remain negative and significant. There are indications of marginally negative returns on Day 3 and marginally positive returns on Day 9 but no other significant results for the remaining 10 days. None of the CARs are statistically significant.

To summarize, although there is a modest and significant negative stock market reaction on the day of the Rana Plaza disaster, the magnitude and significance of the stock market reaction dissipates when the abnormal return on the day after the disaster is also included. Further, there is no evidence of significant negative stock market reaction over an 11-day period. Basically, retailers who source from Bangladesh did not experience negative stock market reaction from the Rana Plaza disaster.

4.2 Stock market reaction to the announcement of the agreements

4.2.1 Stock market reaction to the AFBSB

Table 5 presents the abnormal returns associated with the announcement of the AFBSB accord on May 11, 2013. We report results for the AFBSB signees only, the ABWS signees only, and the combined sample of AFBSB and ABWS firms. Panel A of Table 5 indicates that the abnormal returns for the AFBSB signees are insignificant for all six days of the event period. Further, the median, mean, and percent positive CARs are insignificant for both the two-day period of Day 0 and Day 1, and the six-day period from Day 0 through Day 5.

We also examine the stock market reaction to firms that did not sign the AFBSB but subsequently signed the ABWS. Not signing the AFBSB may convey some information to the stock market about the ABWS signees. Panel A of Table 5 indicates that for ABWS signees, the Day 0 stock market reaction is negative but insignificant and is followed by a significant, positive reaction on Day 1. The Day 1 reaction is similar in terms of the absolute magnitude of the Day 0 reaction. The market reaction over the two-day period of Day 0 and Day 1 is insignificantly different from zero. ABWS signees experience a significant, negative abnormal return on Day 3; the median abnormal return is -0.44% , significantly different from zero at the 5% level. For the six-day period from Day 0 through Day 5, the median, mean, and percent positive CARs are insignificant.

For the combined sample of AFBSB and ABWS signees, there is some evidence of positive abnormal performance on Day 1; the median abnormal return is 0.31% , significantly different from zero at the 10% level. However, the market reaction over the two-day period of Day 0 and Day 1 is insignificantly different from zero. The median, mean, and percent positive CARs from Day 0 through Day 5 are also insignificant.

Panel B of Table 5 presents the results for the adjusted sample. We identify 20 contemporaneous announcements for the 39 sample firms over the six-day period that are unrelated to the Rana Plaza disaster, and estimate abnormal returns by removing the abnormal returns around these announcements. The adjusted results are again substantively similar to those of the unadjusted sample in Panel A of Table 5. For the AFBSB signees and for the combined sample of AFBSB and ABWS firms, none of the abnormal returns or CARs is significant. For ABWS signees, the median abnormal return of -0.62% on Day 0 is significantly different from zero at the 5% level. However, the CAR for the two-day period of Day 0 and Day 1 is insignificant. Considering the six-day period from Day 0 through Day 5, the median CAR is -0.54% , significantly different from zero at the 10% level.

In summary, the AFBSB signees experienced no significant stock market reaction to the announcement of the AFBSB. Although the ABWS signees experienced a small, negative reaction on Day 0, it was offset by a small, positive reaction on Day 1. The combined sample of AFBSB and ABWS signees experienced no significant reaction to the announcement of the AFBSB.

We note that these results reflect the stock market reactions to the announcement of the AFBSB, not to the decision of individual firms to sign the AFBSB. In our sample of 25 AFBSB signees, 14 firms were

part of the original signatories at the time of release on May 11, 2013. The remaining 11 firms signed the AFBSB within the 60-day period following the Rana Plaza disaster but we could only find precise signing dates for 10 of the 11. To examine the stock market reaction to individual firms signing the AFBSB, we designate Day 0 for each firm as the date of their announcement to sign the AFBSB. Table 6 presents the abnormal returns and CARs for the six-day period of Day 0 through Day 5 for both the unadjusted and adjusted sample of AFBSB signees. The only significant result is that in the adjusted sample the percent positive abnormal returns on Day 1 and CAR (0, 1) are significantly different from 50% at the 5% level. However, the median, mean, and percent positive CARs from Day 0 through Day 5 are insignificant. The overall lack of significant stock market reactions to firms signing the AFBSB is similar to that found in Table 5 for AFBSB signees upon the initial release of the AFBSB.

4.2.2 Stock market reaction to the ABWS

Table 7 presents the daily abnormal returns and CARs associated with the announcement of the ABWS agreement on July 8, 2013. Panel A indicates that the abnormal returns for the ABWS signees are positive and statistically significant on Day 0. The median (mean) abnormal return is 0.77% (1.12%), significantly different from zero at the 1% level. Considering the CAR for Day 0 and Day 1, the median (mean) abnormal return is 1.12% (1.42%), significantly different from zero at the 1% (5%) level. 85.71% of ABWS signees experience positive abnormal returns, significantly different from 50% at the 1% level. There is evidence of negative abnormal performance on Day -4; the median abnormal return is -0.49%, significantly different from zero at the 5% level, and 14.29% of firms experience positive abnormal returns, significantly different from 50% at the 1% level. For the six-day period from Day 0 through Day 5, the median, mean, and percent positive CARs are insignificant.

To see whether the ABWS announcement had any effect on the AFBSB signees, we examine the stock market reaction to AFBSB firms at the time of the ABWS release. The abnormal returns for the AFBSB signees are insignificant for all six days of the event period. The CARs for Days (0, 1) and Days (0, 5) are also insignificant. For the combined sample of ABWS and AFBSB signees, there is some evidence of negative abnormal performance on Day 4, which is driven by the negative abnormal performance on Day 4 of ABWS signees. Consistent with the results for the AFBSB signees, the combined sample CARs for Days (0, 1) and Days (0, 5) are both insignificant.

Panel B of Table 7 presents the results for the adjusted sample. We identify 10 contemporaneous announcements for the 39 sample firms over the six-day period that are unrelated to the Rana Plaza disaster, and adjust the abnormal returns by removing abnormal returns around these announcements. The adjusted results are similar to those of the unadjusted sample in Panel A of Table 7. To summarize, ABWS signees experience a small but significant positive stock market reaction on Day 0 but no significant reactions over the six-day period. Further, neither the AFBSB signees nor the combined sample experience significant stock market reactions to the announcement of the ABWS.

The results for the ABWS signees reflect the stock market reactions to both the announcement of the ABWS, and to the decision of individual firms to sign the ABWS. Unlike our sample of AFBSB signees, the 14 ABWS signees in our sample were all original signatories at the time of release on July 8, 2013. No other publicly-traded firms signed the ABWS within the 60-day period following the Rana Plaza disaster. Thus, the stock market reaction to the ABWS announcement is coincident with the decision of the firms to sign the ABWS, and no other analysis is required.

Although the AFBSB and ABWS were formulated by retailers to address the safety issues and working conditions in Bangladeshi garment factories, and there has been much discussion about these agreements in the media, we do not find evidence of significant stock market reactions to these agreements.

4.3 Combined stock market reaction

To provide an overall assessment of the stock market reaction, we combine the abnormal returns for the three periods whose results we have reported above. This includes the 11-day period after the Rana Plaza disaster, and the two six-day periods following the announcements of the AFBSB and ABWS agreements. Together these three periods add up to a 23-day period. Table 8 Panel A presents the results, and Figure 3 plots the time series of the median CAR over the 23-day period for both the unadjusted and adjusted sample. The results indicate that both the unadjusted and adjusted CARs for Day 0 through Day 22 are insignificantly different from zero.

To account for the possibility that other information related to the Rana Plaza disaster may have been released outside of the 23-day period, we also estimate the abnormal returns over a period of 60 consecutive trading days starting from the day of the Rana Plaza accident. Table 8 Panel B indicates that the unadjusted 60-day median CAR is 4.88%, significantly different from zero at the 5% level, and that 71.79% of firms

experience positive abnormal returns, significantly different from 50% at the 1% level. However, when the CAR is adjusted for other contemporaneous announcements unrelated to the Rana Plaza accident, the median and percent positive CAR are both insignificant. Figure 4 plots the time series of the median CAR over the 60-day period for both the unadjusted and adjusted samples.

5. Summary and Implications

5.1 Rana Plaza disaster

Although stock market reaction to retailers on the day of the Rana Plaza disaster is negative and significant, its magnitude and significance dissipates when considering the following day's stock market reaction. There is no evidence of significant stock market reaction over 11 trading days (approximately two weeks in calendar time) after the Rana Plaza disaster. Further, for the 60 trading days (approximately three months in calendar time) after the Rana Plaza disaster, the stock market reaction adjusted for other contemporaneous announcements unrelated to the Rana Plaza disaster is statistically insignificant. Basically, retailers who source from Bangladesh did not experience any significant negative economic impact from the Rana Plaza disaster. These results question the common assumption that market forces alone might prompt firms to shift sourcing from low-cost countries to higher-cost countries.

As discussed in our development of H1, one reason that we expected a negative stock market reaction to the Rana Plaza disaster was the potential for supply disruptions. The insignificant reaction suggests that investors do not see the Rana Plaza disaster as significantly disrupting the apparel supply chains of retailers that source from Bangladesh. This result contrasts with the work of Hendricks and Singhal (2003, 2005b) who demonstrate that the stock market reaction to supply chain disruptions is significantly negative. Although some supply disruptions undoubtedly resulted from the disaster, their impact was probably limited due to the nature of the apparel retail industry. Clothing retailers carry large assortments of garments. For each potentially stocked-out item, alternative brands, styles, and colors might be acceptable to consumers. Another mitigating factor on the impact of disruptions is the role of supply chain inventories and slack. If inventories for the affected items were sufficient at the time of the disaster, disruption effects would be reduced. Further, of the more than 5,000 garment factories in Bangladesh, only a few were housed in Rana Plaza. There were also no long-lasting, industry-wide strikes or factory closures in the aftermath of the Rana Plaza disaster. According to data released by the Bangladesh Export Promotion Bureau,

garment exports climbed 24% during the months of July through September in 2013 when compared to the same time period in 2012. It seems that the Rana Plaza disaster neither disrupted the flow of garments from Bangladesh nor did it stop the expansion of the Bangladeshi garment industry.

We searched corporate press releases and the global business press for any reports of supply disruptions from Bangladesh for our sample retailers during the six-month period subsequent to the disaster. Sources include *The Wall Street Journal*, *Herald Sun*, *Globe and Mail*, *Toronto Star*, *Financial Times*, *International Herald Tribune*, *The Daily Telegraph*, *The Times*, and both the US and foreign versions of *Dow Jones News Service* and *PR Newswire*. We found no reports of supply disruptions.

The second reason we expected a negative stock market reaction to the Rana Plaza disaster was the potential for reputational damage. Our finding of an insignificant stock market reaction to the Rana Plaza disaster suggests that investors do not hold retailers responsible for the Rana Plaza disaster. Although retailer reputations were in jeopardy due to the many criticisms in the aftermath of the Rana Plaza disaster, the accident investigation concluded that factors beyond retailer control constituted the direct causes. These factors include shoddy construction of the building using poor quality material; unauthorized and illegal addition of floors in the building; poor enforcement of building codes and laws; and forcing workers back to work under the threat of lost wages and/or discharge even after visible cracks appeared in the structure of the building (Yardley 2013). Retailers had no ownership stake in the Rana Plaza suppliers, nor did they directly manage the construction of the Rana Plaza building or the daily operations of the factories in the building. Further, the apparel supply chains in Bangladesh are heavily fragmented, intermediated, and opaque, which obscures the responsibility associated with retailers. The stock market reaction suggests that investors do not hold retailers responsible for the Rana Plaza disaster and also do not see much, if any, reputational damage from the incident.

5.2 AFBSB and ABWS agreements

Retailers responded to the Rana Plaza disaster by offering a coordinated response to address safety and working conditions in Bangladeshi garment factories. They developed and implemented two different accords – the AFBSB and the ABWS. Analysis of the stock market reactions to the announcements of these two agreements finds no evidence of significant reactions. This could indicate that the investors view these agreements as not that different from what retailers were already doing to ensure safety and improve

working conditions in Bangladeshi garment factories. Even before the Rana Plaza disaster, most large retailers had inspection systems to monitor and evaluate safety and working conditions in factories from which they sourced. However, there was little coordination of these activities across retailers, and factories with multiple customers could be subjected to multiple inspections with different standards. The lack of a significant stock market reaction calls into question the assumption that market forces alone might prompt firms to significantly invest in improving working conditions in low-cost countries.

The AFBSB and ABWS accords centralize inspection and monitoring across retailers by using a uniform set of standards and a common inspection team, thereby avoiding duplication. It can also lead to better coordination, information sharing, and visibility. Although this should lead to cost savings in inspection and monitoring activities, the lack of stock market reaction suggests that the savings are not that substantial. For example, the amount contributed by each member of the ABWS is based on the amount of production the company has in Bangladesh, with a cap of \$1 million per year. Given the average retailer in our sample has median annual sales of \$11.4 billion, the maximum of \$1 million per year is not a large investment, and costs savings from joining the alliance are likely not significant.

5.3 Implications

The results of our analysis of the stock market reaction to retailers who source from Bangladesh have important implications for retailers, garment factory owners in Bangladesh, NGOs, policymakers including the Bangladeshi government, as well as researchers. We next discuss these implications.

5.3.1 Retailers

From a retailer perspective, since negative economic impacts from the Rana Plaza disaster were insignificant, they have limited economic incentive to move sourcing out of Bangladesh to other low-cost countries such as Cambodia and China, both of which have well-developed garment industries, let alone to higher-cost countries in North America or Europe. This evidence is counter to the widely-held assumption that the risks of sourcing in developing, low-cost countries might lead to increased sourcing in developed, high-cost countries. There is also no evidence to suggest that customers are boycotting garments made in Bangladesh. This is consistent with the arguments of Diermeier (2012) that consumer boycotts are difficult to motivate and organize. On the contrary, as mentioned earlier, garment exports from Bangladesh actually increased in the months after the Rana Plaza disaster. Further, the time and effort to shift \$20 billion of

exports from Bangladesh to another country is not trivial (Fahey and D’Innocenzio 2013). Bangladesh is currently considered the lowest-cost garment manufacturing country with significant manufacturing capabilities and capacity. It would require significant resources on the part of the retailers to find countries that have the capacity to manufacture garments in such high volume. Finally, even if retailers move production out of Bangladesh to other low-cost countries, several of these countries might have working conditions issues similar to Bangladesh.

Our findings are based on a single, large-scale disaster. If disasters of similar scale were to reoccur in the future, the stock market reactions to retailers could be different in a number of ways. First, investors might assign more responsibility to retailers if such disasters were to repeat after retailer pledges to improve conditions. Second, repeated disasters might prompt consumer activism and boycotts. Third, governments in developed economies might respond with trade sanctions or suspensions of trade privileges that could lead to major supply chain disruptions. The chance of a reoccurrence, along with ethical and moral issues, might prompt some retailers to move their sourcing out of Bangladesh. For example, Nike limited its use of factories in Bangladesh because of concerns about working conditions (Banjo 2014) and Disney discontinued sourcing Bangladesh and other poor-governance countries (Barrie 2014). This is a choice that individual retailers have to make as they balance risk, ethical, and moral considerations against competitive issues such as pricing and availability.

Another implication of our analysis is that retailers should be cautious of taking direct responsibility for the safety in Bangladeshi garment factories. Direct responsibility exposes retailers to greater liability and reputation issues. It can also create moral hazards such that factory owners might shirk their responsibility toward safety, not invest in factory remediation, and thereby ultimately increase the costs and risk faced by retailers. Bangladesh suffers from a high level of government corruption as measured by Transparency International (2013) and the World Bank (2013). At least 10 percent of the parliament members in Bangladesh are garment factory owners. These members use their influence to keep wages low and maintain political clout for favorable treatment of the garment industry including lax safety regulations (Chalmers 2013). If retailers assume direct responsibility in such corrupt and politically connected environments, they can be exposed to unknown risks. Finally, if retailers are directly responsible for factory safety, they will need to create some form of a consortium to collectively address safety issues.

Creating and managing such a consortium can cause challenging competitive issues as well as issues on how to share both the cost of ensuring safety and the responsibility for any future accidents that might occur. Given the above, a reasonable course of action for retailers to improve safety seems to be to continue their present course: monitor the workplace through inspections; identify deficiencies; give factory owners time and support to fix the problems; and use the threat of taking away business to ensure compliance.

5.3.2 Factory owners

The role that factory owners play in workplace safety improvements is paramount. As found by the accident investigation, the Rana Plaza building owner and factory owners were the responsible parties (Yardley 2013). Some factory owners are demanding that retailers pay higher prices, and stop squeezing margins. They claim their margins are insufficient to permit safe workplaces, forcing them to make cuts that threaten safety (Srivastava 2013, Chalmers 2013). However, if retailers pay higher prices, it is not clear that the factory owners will use the increased margins to improve workplace safety. Further, the claims of insufficient margins appear shallow in light of profitability analyses of 34 publicly traded firms that comprise the Bangladeshi textile sector listed on the Dhaka Stock Exchange. Based on information collected from the Dhaka Stock Exchange website, the median return on assets (ROA) of these 34 firms in the 2012 fiscal year is 6.88%. In contrast, the median ROA of the 39 retailers in our sample in the 2012 fiscal year is 5.15%. Our review of the websites of the 34 Bangladeshi textile firms indicates that eight firms are primarily ready-made garment makers. The median ROA of these eight firms is 8.24% in the 2012 fiscal year. In contrast, the median ROA of the 11 firms in our sample in apparel retailing (SIC code 56) is 7.93%. Based on this limited analysis, it seems that the large garment manufacturers in Bangladesh are profitable, and their profitability is similar to the retailers that source from them. In addition, the rapid growth of the Bangladeshi garment industry in the last few years shows no sign of slowing. Such rapid growth would be unlikely if the industry is not profitable.

Beyond the moral and ethical incentives for factory owners to improve workplace safety, economic incentives include avoiding production losses, reducing liabilities, and decreasing reputational risks. Further, more accidents and loss of life will undoubtedly slow industry growth as retailers may eventually shift production to other countries. Given the above, factory owners have powerful incentives to take workplace safety seriously and need to take actions to ensure workplace safety.

5.3.3 NGOs

Our results may be surprising to many, especially advocates of corporate social responsibility. What our research highlights is that capital market forces alone are insufficient to prevent tragedies such as the Rana Plaza disaster, or to motivate improved labor conditions in developing countries. While managers should weigh ethics and their moral obligation in addition to financial considerations, it is doubtful that firms can affect the needed changes without participation by non-market forces. This adds evidence to the recent arguments of Locke (2013) about the limited power of buyers or NGOs to affect widespread change in supplier labor conditions. He advocates for more active involvement by governments, and coordinated efforts by buyers, NGOs, and governments.

NGOs can play an important role in ensuring safety, good working conditions, and fair treatment of workers across supply chains. In several past instances, their efforts highlighted lapses in social performance across supply chains and led companies to address these issues. However, considering the reaction of investors to the Rana Plaza disaster, the approach used by many NGOs to shame and guilt retailers into accepting responsibility for the safety issues in the Bangladesh garment industry might be misplaced. Requiring retailers to assume direct responsibility for factory safety could lead more retailers to leave Bangladesh since the possibility of future disasters cannot be ruled out, and the possibility of being tainted in the future is not negligible. If retailers exit on a large scale, garment workers in Bangladesh stand to lose. Rather than scapegoating retailers, the NGOs can acknowledge that the blame belongs with the factory owners that neglect safety and with lax governmental enforcement, and then work cooperatively with various stakeholders to proactively address those issues.

5.3.4 Policymakers

The regulatory laws in Bangladesh are weak and the implementation is uneven and inconsistent (Labowitz and Baumann-Pauly 2014). Moreover, the political connections of factory owners, widespread corruption, and a justice system that moves slowly allows factory owners to avoid responsibility for workplace accidents. In her book that chronicles the evolution of the garment industry, Rivoli (2009) notes that in each of the successive low-cost regions to which the garment industry has relocated (e.g., England, US, Latin America, Asia), large-scale tragedies and worker exploitation have been unfortunate consequences. In each instance, market forces alone were insufficient to protect workers, and it was not

until governments established and enforced regulatory standards, and factory owners complied with them, that such problems were substantially reduced. Despite the arguments of Rivoli (2009), it is evident that the popular press and many stakeholders placed much of the blame for the Rana Plaza disaster on retailers rather than regulators, suggesting that market forces can rectify the problem. Our finding of an insignificant stock market reaction to the Rana Plaza disaster lends rigorous, empirical evidence to the claim of Rivoli (2009) that market forces alone are insufficient. It is critical for the Bangladeshi government to set clear and comprehensive regulatory guidelines, and consistently enforce them.

We note that the Rana Plaza disaster comprises four key elements that might be used to assess the applicability of our findings to other incidents: 1) the event has major, negative social impact with extensive media coverage; 2) buyers are largely located in developed economies; 3) the supply base is fragmented and opaque, and largely located in developing economies; and 4) consumer backlash is limited. To the extent that other events share these elements, insignificant stock market reactions might be expected. The prevalence of these four factors in today's globalized supply chains should prompt action by policymakers to devise and implement strategies aimed at reducing the likelihood of such events.

The governments of buying firms in more developed economies can also play a role by altering trade agreements to pressure Bangladesh to improve worker rights and safety. For example, the US government suspended some trade privileges for Bangladesh as a result of the Rana Plaza disaster (Associated Press 2013), and the EU entered a sustainability compact with the Bangladeshi government that includes time-based actions to improve labor rights and factory safety (European Commission 2013). Without such governmental policies, it seems unlikely Bangladeshi worker safety will substantively improve in the short term.

5.3.5 Researchers

Our results also have implications for the growing body of academic research on responsible sourcing, supplier risk, and offshoring/reshoring decisions. It is a common assumption in analytical models that customer willingness-to-purchase is partially contingent on the social performance of suppliers; i.e., consumers might boycott sales if suppliers violate social norms, or poor social performance of suppliers imposes significant costs on buyers. For example, Guo et al. (2013) model the demand disruption effects resulting from supplier social violations, and Chen and Baddam (2013) model the relationship between the

likelihood of supplier socially irresponsible events and consumer willingness-to-pay for ethically produced goods. If consumer reaction is sufficiently negative, it would depress firm cash flows, reducing stock prices. We find that retailers who source from Bangladesh did not experience any significant negative economic impact from the Rana Plaza disaster. This empirical evidence can serve as a benchmark for researchers that model financial impacts of supplier social disasters. Rather than modeling firm sourcing decisions as driven by a direct cause-and-effect between supplier incidents and consumer purchases, more nuanced models incorporating regulatory actions and coordinated stakeholder efforts are likely more appropriate. Our finding also lends support to the offshoring/reshoring model developed by Gray et al. (2015). A key driver in their model is the gap between desired and actual offshore supplier performance, which is widened by unanticipated supplier problems. As they discuss, the current major driver of that gap is cost rather than social or environmental issues.

5.4 Limitations and Future Research

One limitation of our study is that we study a single event, the Rana Plaza disaster. Although the event is infamous and had repercussions throughout the global garment industry, generalizing our results to other industries or contexts must be done with caution. For example, the stock market reaction might have been quite different if the Rana Plaza factories all supplied a single retailer, or if Bangladeshi factories were the sole global suppliers of ready-made garments. Studying the effects of similar events perhaps in different industries or contexts would aid in better understanding the risks and benefits of sourcing in developing countries. Further, although our sample of retailers is international, it is restricted to primarily North American and European firms. Buying firms in developing economies might be subject to different reactions from NGOs, the public, and other stakeholders.

In this study, we do not directly measure supply disruptions or reputational impacts but instead we infer them from the stock market reaction. Future research could directly explore the magnitude and significance of any changes to disruptions or reputations from tragedies like Rana Plaza, or from events such as major reshoring initiatives. Such an approach would require before- and after-assessments of disruptions or reputations, and would yield insights into the exact mechanisms through which such events impact firms. Aside from disruptions and reputations, sourcing in low-cost countries such as Bangladesh of course also impacts the natural environment. It would be interesting to assess the impact of sourcing decisions to

carbon footprints or other environmental measures. Environmental impacts could be translated into economic impacts by applying cost-of-carbon or other conversions, and it would be possible to estimate the carbon cost sufficient to materially impact a firm's economic bottom line. Future research could also study other measures of financial impact including changes to profitability or operating performance.

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Table 1. Comparison of the key features of the Accord on Fire and Building Safety in Bangladesh (AFBSB) and the Alliance for Bangladesh Worker Safety (ABWS) agreements.

Item	Accord on Fire and Building Safety in Bangladesh (AFBSB)	Alliance for Bangladesh Worker Safety (ABWS)
Member firms	Primarily European firms	North American firms only
Governing Committee	Equal representation by 3 company signees and 3 trade union signees, plus a chair from the International Labor Organization	Equal representation by 4 company signees and 4 other stakeholders (NGO, industry), plus former US State Department official as chair
Inspections	Chief inspector runs independent inspection program; full inspections for largest 65% of factories and partial inspections at remaining	Committee of Experts sets standards and approves inspectors; full inspections for all factories within the first year
Dispute resolution	Disputes settled by majority vote; appeals to a binding arbitration process; award enforceable in company's home country courts	Not addressed
Subcontractors	Not addressed	Share best practices to prohibit unauthorized subcontracting
Sourcing requirement	Must maintain sourcing levels in Bangladesh for at least two years	Not addressed
Garment prices	Prices should ensure financial feasibility for factories to maintain safe workplaces	Not addressed
Financial assistance	Option to provide loans, joint investments, business incentives, or direct payment for remediation efforts	Option to provide loans, supplemental contributions, or in-kind contributions for remediation efforts
Membership fees	Dues proportional to Bangladesh garment purchases, maximum of \$500,000 per year	Dues proportional to Bangladesh garment purchases, maximum of \$1,000,000 per year

Table 2. Names and headquarters of publicly traded firms that signed the Accord on Fire and Building Safety In Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements within the 60-day period following the Rana Plaza disaster on April 24, 2013.

<i>Panel A. Accord on Fire and Building Safety in Bangladesh (AFBSB)</i>			
Company	Headquarters	Company	Headquarters
Abercrombie & Fitch	US	Loblaw	Canada
American Eagle Outfitters	US	Marks & Spencer (M&S)	UK
Carrefour	France	Metro Group	Germany
Charles Vögele	Switzerland	Mothercare	UK
Dansk Supermarked	Denmark	N Brown Group	UK
Debenhams	UK	Next	UK
Esprit	Hong Kong	Primark	UK
Hemtex AB	Sweden	Puma	Germany
Hennes & Mauritz (H&M)	Sweden	PVH	US
IC Companys	Denmark	Sainsbury's	UK
Inditex	Spain	Stockmann	Finland
KappAhl	Sweden	Tesco	UK
Kmart & Target Australia	Australia		
<i>Panel B. Alliance for Bangladesh Worker Safety (ABWS)</i>			
Company	Headquarters	Company	Headquarters
Canadian Tire Corporation	Canada	Nordstrom	US
Carter's	US	Sears Holdings Corporation	US
Gap Inc.	US	Target Corporation	US
Hudson's Bay Company	Canada	The Children's Place	US
J.C. Penney Company	US	The Jones Group	US
Kohl's Department Stores	US	VF Corporation	US
Macy's	US	Wal-Mart Stores	US

Table 3. Descriptive statistics of the sample of 39 firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements.

Panel A. Top four corporate headquarters countries among sample firms

Country	Occurrences (%)
United States	15 (38.5%)
United Kingdom	7 (17.9%)
Sweden	3 (7.7%)
Canada	3 (7.7%)

Panel B. Top two primary SIC codes among sample firms

SIC	Industry Description	Occurrences (%)
53	Retail - General Merchandise	13 (33.3%)
56	Retail - Apparel	11 (28.2%)

Panel C. Sample firm statistics at the end of the 2012 fiscal year

	Market Value (\$ million)	Total Assets (\$ million)	Sales (\$ million)	Net Income (\$ million)	Employees (000s)
Mean	20,614	29,088	40,950	1,893	157.7
Median	8,667	8,089	11,427	434	71.5
Std Dev	40,756	72,895	90,983	4,972	360.5

Table 4. Abnormal returns for the unadjusted and adjusted sample of firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements; Event Day 0 is the date of the Rana Plaza disaster on April 24, 2013.

<i>Panel A. Unadjusted sample</i>							
Event Day(s)	<i>N</i>	Median	<i>Z</i> ^a	Mean	<i>t</i>	% Negative	<i>Z</i> ^b
0	39	-0.40%	-2.15 **	-0.27%	-0.68	66.67%	2.08 **
1	39	0.16%	1.36	0.49%	1.23	43.59%	-0.80
2	39	0.13%	0.19	0.18%	0.46	43.59%	-0.80
3	39	-0.28%	-1.52	-0.29%	-0.73	58.97%	1.12
4	39	0.15%	1.03	0.27%	0.68	41.03%	-1.12
5	39	0.09%	0.07	-0.02%	-0.06	48.72%	-0.16
6	39	0.05%	0.65	0.22%	0.55	46.15%	-0.48
7	39	0.03%	0.83	0.25%	0.63	48.72%	-0.16
8	39	-0.54%	-1.40	-0.24%	-0.60	64.10%	1.76 *
9	39	0.52%	1.62	0.35%	0.88	35.90%	-1.76 *
10	39	-0.14%	-1.11	-0.05%	-0.12	61.54%	1.44
(0, 1)	39	-0.21%	0.07	0.22%	0.38	51.28%	0.16
(0, 5)	39	0.52%	0.62	0.35%	0.36	46.15%	-0.48
(6, 10)	39	0.00%	0.53	0.54%	0.60	48.72%	-0.16
(0, 10)	39	0.94%	1.10	0.89%	0.68	43.59%	-0.80
<i>Panel B. Adjusted sample</i>							
Event Day(s)	<i>N</i>	Median	<i>Z</i> ^a	Mean	<i>t</i>	% Negative	<i>Z</i> ^b
0	29	-0.57%	-2.91 ***	-0.62%	-1.57	72.41%	2.41 ***
1	29	0.06%	0.65	0.22%	0.56	48.28%	-0.19
2	31	0.19%	0.78	0.08%	0.20	38.71%	-1.26
3	30	-0.26%	-1.74 *	-0.40%	-1.00	60.00%	1.10
4	29	0.18%	1.61	0.54%	1.37	37.93%	-1.30
5	30	-0.21%	-1.06	-0.36%	-0.90	56.67%	0.73
6	34	0.13%	0.64	0.16%	0.41	47.06%	-0.34
7	34	0.00%	0.67	0.25%	0.64	50.00%	0.00
8	32	-0.42%	-0.57	-0.01%	-0.03	59.38%	1.06
9	33	0.52%	1.99 **	0.49%	1.23	33.33%	-1.91 *
10	31	-0.14%	-1.26	-0.35%	-0.88	61.29%	1.26
(0, 1)	32	-0.64%	-1.01	-0.36%	-0.65	56.25%	0.71
(0, 5)	38	-0.23%	-0.91	-0.42%	-0.44	57.89%	0.97
(6, 10)	39	0.40%	0.71	0.49%	0.55	46.15%	-0.48
(0, 10)	39	0.03%	0.53	0.08%	0.06	48.72%	-0.16

Notes:

^a *Z*-statistics for medians are obtained using Wilcoxon signed-rank tests

^b *Z*-statistics for % negatives are obtained using binomial sign tests

All tests are two-tailed: *** $p \leq 0.01$; ** $p \leq 0.05$; * $p \leq 0.10$

Table 5. Abnormal returns for the unadjusted and adjusted sample of firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements; Event Day 0 is the first trading day after the AFBSB initial announcement on May 11, 2013.

Panel A. Unadjusted sample												
Event Day(s)	AFBSB Signees				ABWS Signees				AFBSB and ABWS Signees			
	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b
0	25	-0.64%	0.23%	40.00%	14	-0.58%	-0.23%	35.71%	39	-0.62%	0.06%	38.46%
1	25	-0.05%	0.15%	48.00%	14	0.52% ***	0.64%	85.71% ***	39	0.31% *	0.32%	61.54%
2	25	-0.46%	-0.33%	44.00%	14	0.38%	0.25%	57.14%	39	-0.13%	-0.12%	48.72%
3	25	0.16%	-0.14%	56.00%	14	-0.44% **	-0.24%	14.29% ***	39	-0.19%	-0.18%	41.03%
4	25	-0.14%	-0.06%	48.00%	14	0.04%	-0.57%	57.14%	39	0.01%	-0.24%	51.28%
5	25	-0.34%	0.16%	40.00%	14	-0.18%	0.28%	42.86%	39	-0.25%	0.21%	41.03%
(0, 1)	25	-0.43%	0.37%	48.00%	14	-0.06%	0.41%	42.86%	39	-0.10%	0.39%	46.15%
(0, 5)	25	-0.24%	0.01%	44.00%	14	-0.07%	0.13%	50.00%	39	-0.24%	0.05%	46.15%

Panel B. Adjusted sample												
Event Day(s)	AFBSB Signees				ABWS Signees				AFBSB and ABWS Signees			
	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b
0	19	-0.49%	0.36%	42.11%	11	-0.62% **	-0.61%	27.27%	30	-0.58%	0.01%	36.67%
1	20	-0.14%	-0.05%	40.00%	9	0.74% **	0.75% *	88.89% **	29	0.29%	0.20%	55.17%
2	21	-0.75%	-0.68%	33.33%	9	-0.17%	-0.19%	44.44%	30	-0.57%	-0.53%	36.67%
3	21	0.05%	-0.18%	52.38%	9	-0.48% **	-0.63%	11.11% **	30	-0.38%	-0.31%	40.00%
4	22	0.02%	0.07%	50.00%	10	0.39%	0.15%	80.00% *	32	0.23%	0.10%	59.38%
5	22	-0.36%	0.00%	40.91%	11	-0.11%	0.00%	45.45%	33	-0.22%	0.00%	42.42%
(0, 1)	21	-0.50%	0.28%	42.86%	11	-0.10%	0.00%	36.36%	32	-0.29%	0.19%	40.63%
(0, 5)	24	-0.83%	-0.44%	37.50%	12	-0.54% *	-0.49%	33.33%	36	-0.66%	-0.45%	36.11%

Notes: ^a Z-statistics for medians are obtained using Wilcoxon signed-rank tests

^b Z-statistics for % positives are obtained using binomial sign tests

All tests are two-tailed: *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$

Table 6. Abnormal returns for the unadjusted and adjusted sample of firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB); for each firm, Event Day 0 is the date of the initial announcement of its signing.

<i>Panel A. Unadjusted sample</i>							
Event Day(s)	<i>N</i>	Median	<i>Z</i> ^a	Mean	<i>t</i>	% Positive	<i>Z</i> ^b
0	24	-0.47%	-1.52	-0.21%	-0.52	37.50%	-1.22
1	24	-0.23%	-0.75	-0.07%	-0.18	33.33%	-1.63
2	24	-0.49%	-1.04	-0.43%	-1.09	41.67%	-0.82
3	24	0.37%	1.28	0.44%	1.10	66.67%	1.63
4	24	0.25%	0.39	0.07%	0.17	58.33%	0.82
5	24	-0.38%	-0.66	0.11%	0.28	41.67%	-0.82
(0, 1)	24	-0.52%	-1.18	-0.28%	-0.49	33.33%	-1.63
(0, 5)	24	-0.49%	-0.72	-0.10%	-0.10	41.67%	-0.82
<i>Panel B. Adjusted sample</i>							
Event Day(s)	<i>N</i>	Median	<i>Z</i> ^a	Mean	<i>t</i>	% Positive	<i>Z</i> ^b
0	20	-0.26%	-1.02	-0.05%	-0.13	40.00%	-0.89
1	20	-0.40%	-1.39	-0.31%	-0.77	25.00%	-2.24 **
2	22	-0.82%	-1.36	-0.57%	-1.43	36.36%	-1.28
3	21	0.37%	0.77	0.34%	0.85	61.90%	1.09
4	22	0.28%	0.75	0.20%	0.49	59.09%	0.85
5	20	-0.35%	-0.68	-0.05%	-0.14	45.00%	-0.45
(0, 1)	21	-0.54%	-1.38	-0.34%	-0.61	28.57%	-1.96 **
(0, 5)	23	-0.84%	-1.35	-0.41%	-0.42	39.13%	-1.04

Notes: ^a *Z*-statistics for medians are obtained using Wilcoxon signed-rank tests

^b *Z*-statistics for % negatives are obtained using binomial sign tests

All tests are two-tailed: *** $p \leq 0.01$; ** $p \leq 0.05$; * $p \leq 0.10$

Table 7. Abnormal returns for the unadjusted and adjusted sample of firms that signed the Alliance for Bangladesh Worker Safety (ABWS) or the Accord on Fire and Building Safety in Bangladesh (AFBSB) agreements; Event Day 0 is the date of the ABWS initial announcement on July 8, 2013.

Panel A. Unadjusted sample												
Event Day(s)	ABWS Signees				AFBSB Signees				AFBSB and ABWS Signees			
	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b
0	14	0.77% ***	1.12% ***	71.43%	25	-0.08%	-0.43%	44.00%	39	0.07%	0.12%	53.85%
1	14	0.11%	0.30%	64.29%	25	-0.29%	-0.03%	32.00% *	39	-0.07%	0.09%	43.59%
2	14	0.19%	0.23%	57.14%	25	-0.07%	0.01%	40.00%	39	-0.05%	0.09%	46.15%
3	14	-0.24%	-0.24%	35.71%	25	0.02%	0.07%	56.00%	39	-0.09%	-0.04%	48.72%
4	14	-0.49% **	-0.60%	14.29% ***	25	-0.05%	-0.29%	44.00%	39	-0.37% **	-0.40%	33.33% **
5	14	-0.13%	0.16%	42.86%	25	0.25%	0.32%	60.00%	39	0.11%	0.26%	53.85%
(0, 1)	14	1.12% ***	1.42% **	85.71% ***	25	-0.37%	-0.46%	32.00% *	39	0.02%	0.21%	51.28%
(0, 5)	14	0.74%	0.97%	64.29%	25	0.05%	-0.35%	52.00%	39	0.35%	0.12%	56.41%

Panel B. Adjusted sample												
Event Day(s)	ABWS Signees				AFBSB Signees				AFBSB and ABWS Signees			
	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b	N	Median ^a	Mean	% Positive ^b
0	13	0.64% **	0.68%	69.23%	23	-0.04%	-0.39%	47.83%	36	0.08%	0.00%	55.56%
1	13	0.09%	0.23%	61.54%	23	-0.24%	0.09%	34.78%	36	-0.04%	0.14%	44.44%
2	13	0.15%	0.23%	53.85%	22	-0.06%	0.04%	45.45%	35	-0.05%	0.11%	48.57%
3	13	-0.27%	-0.28%	30.77%	21	0.02%	0.11%	57.14%	34	-0.09%	-0.04%	47.06%
4	13	-0.61% ***	-0.67%	7.69% ***	19	-0.05%	-0.31%	42.11%	32	-0.46% **	-0.46%	28.13% ***
5	14	-0.13%	0.16%	42.86%	21	0.25%	0.16%	61.90%	35	0.11%	0.16%	54.29%
(0, 1)	13	1.06% ***	0.91%	84.62% ***	23	-0.24%	-0.30%	34.78%	36	0.03%	0.14%	52.78%
(0, 5)	14	0.12%	0.33%	50.00%	25	0.05%	-0.25%	52.00%	39	0.05%	-0.04%	51.28%

Notes: ^a Z-statistics for medians are obtained using Wilcoxon signed-rank tests

^b Z-statistics for % positives are obtained using binomial sign tests

All tests are two-tailed: *** $p \leq .01$; ** $p \leq 0.05$; * $p \leq 0.10$

Table 8. Cumulative abnormal returns for the unadjusted and adjusted sample of firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements; Event Day 0 is the date of the Rana Plaza disaster on April 24, 2013.

<i>Panel A.</i> Three event periods combined (23 days)			
Event Days	AFBSB and ABWS Signees		
	Median ^a	Mean	% Positive ^b
Unadjusted sample	0.51%	1.07%	56.41%
Adjusted sample	-0.20%	-0.38%	48.72%

<i>Panel B.</i> 60 consecutive trading days			
Event Days	AFBSB and ABWS Signees		
	Median ^a	Mean	% Positive ^b
Unadjusted sample	4.88% **	4.45%	71.79% ***
Adjusted sample	1.72%	1.23%	58.97%

Notes: ^a Z-statistics for medians are obtained using Wilcoxon signed-rank tests

^b Z-statistics for % positives are obtained using binomial sign tests

All tests are two-tailed: *** $p \leq .01$; ** $p \leq 0.05$; * $p \leq 0.10$

		AFBSB or ABWS Signatory Firm?	
		Yes	No
Sourcing in Bangladesh?	Yes	1. Most affected	2. Less affected
	No	3. Less affected and unlikely to occur	4. Least affected

Figure 1. Four quadrant matrix segmenting retailers based on whether they signed the AFBSB or the ABWS, and whether they source from Bangladesh.

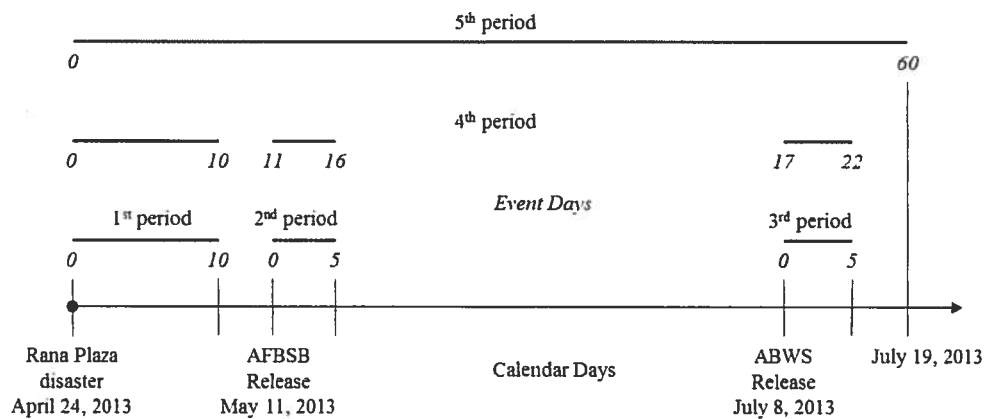


Figure 2. Timeline illustrating the relationship of calendar days and event days for the five event periods considered; 1st period comprises 11 trading days to capture the immediate effects of the Rana Plaza disaster; 2nd and 3rd periods comprise six trading days each to capture the effects of the AFBSB and ABWS announcements, respectively; 4th period aggregates periods 1, 2, and 3 and comprises 23 trading days; 5th period comprises the 60 consecutive trading days subsequent to the Rana Plaza disaster; US calendar dates are shown.

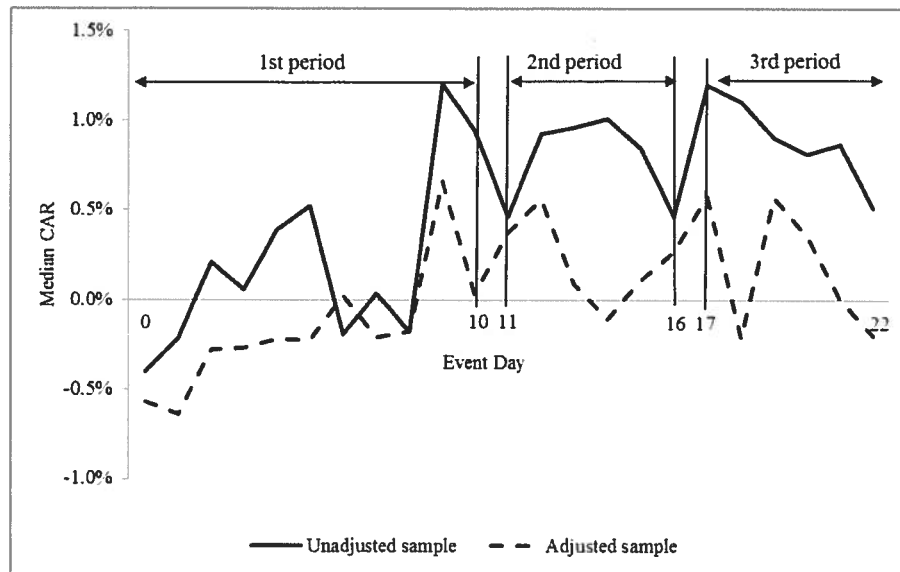


Figure 3. Median cumulative abnormal returns for the unadjusted and adjusted sample of firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements for the 4th event period aggregating periods 1, 2, and 3, and comprising 23 event days in total; Event Day 0 is the date of the Rana Plaza disaster on April 24, 2013.

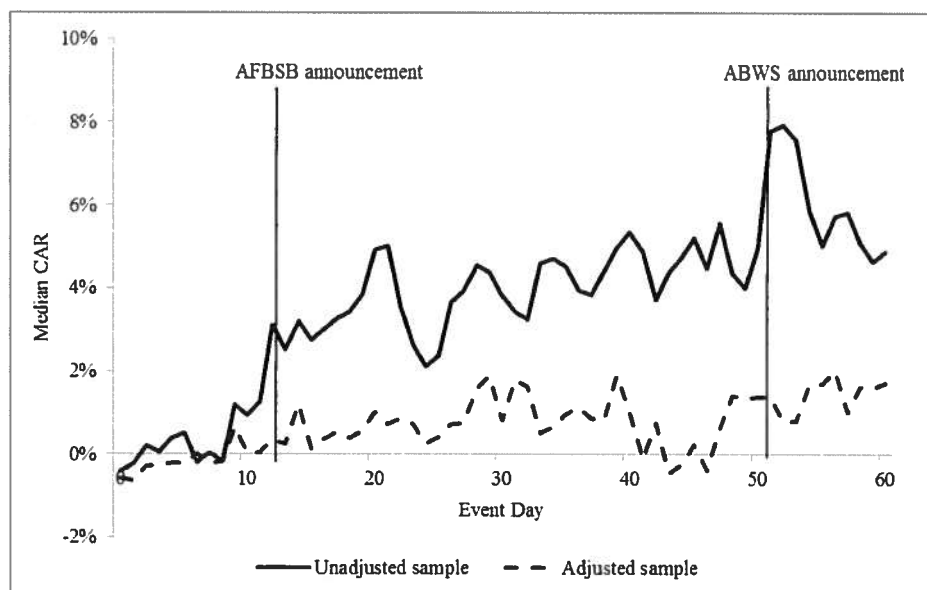


Figure 4. Median cumulative abnormal returns for the unadjusted and adjusted sample of firms that signed the Accord on Fire and Building Safety in Bangladesh (AFBSB) or the Alliance for Bangladesh Worker Safety (ABWS) agreements for the 5th event period comprising 60 consecutive trading days subsequent to the Rana Plaza disaster on April 24, 2013.

Appendix A

The market model posits a linear relationship between the stock return and the market portfolio return over a given time period as:

$$R_{it} = \alpha_i + \beta_i R_{itm} + \varepsilon_{it} \quad (A1)$$

where R_{it} is the Day t return of stock i , R_{itm} is the Day t market return for stock i , α_i is the intercept of the relationship for stock i , β_i is the slope of the relationship for stock i with respect to its market return, and ε_{it} is the error term for stock i on Day t . The portion of stock i 's return attributed to the movements of its market is represented by the term $\beta_i R_{itm}$. The error term ε_{it} is the portion of the return that is unexplained by market movements, and therefore captures the effect of firm-specific information. Since stocks of our sample firms are traded in a variety of markets, we use the dominant market index in each country to generate R_{itm} for stock i . For example, the market index used for US firms is the S&P 500 index and the market index used for UK firms is the FTSE 100 index.

To compute each sample firm's expected return, we estimate $\hat{\alpha}_i$ and $\hat{\beta}_i$ using ordinary least squares regression over the estimation period of 200 trading days. Consistent with many event studies, we begin the estimation period 210 trading days prior to the Rana Plaza disaster on April 24, 2013, and end it 11 trading days prior. Firms must have a minimum of 40 stock returns during the 200-day estimation period to estimate equation (A1).

The abnormal return A_{it} for firm i on Day t is computed as the difference between the actual and the expected return:

$$A_{it} = R_{it} - (\hat{\alpha}_i + \hat{\beta}_i R_{itm}) \quad (A2)$$

The Day t mean abnormal return is given by:

$$\bar{A}_t = \sum_{i=1}^N \frac{A_{it}}{N} \quad (A3)$$

where N is the number of firms in the sample on Day t .

Since we are estimating the abnormal returns for multiple firms due to events on the same calendar dates, the event dates of our sample firms are "clustered". Clustering can cause positive cross-sectional dependency across the sample firms. If cross-sectional dependence is not addressed, we may underestimate the standard deviation, and therefore overestimate the magnitude of the test statistics. To control for

clustering, we estimate the standard deviation using the clustered version test advocated by Brown and Warner (1985). We first compute $\bar{A}$, the mean abnormal return for the 200-day estimation period as:

$$\bar{A} = \frac{\sum_{t=-210}^{t=-11} \bar{A}_t}{200} \quad (A4)$$

The standard deviation is then estimated from the time-series of mean daily abnormal returns for the 200-day estimation period as:

$$\hat{S}(\bar{A}_t) = \sqrt{\sum_{t=-210}^{t=-11} (\bar{A}_t - \bar{A})^2 / 199} \quad (A5)$$

The Day t test statistic TS_t is calculated as:

$$TS_t = \bar{A}_t / \hat{S}(\bar{A}_t) \quad (A6)$$

We use t -tests to determine the statistical significance of the mean abnormal returns. For a given time period (t_1, t_2) , the cumulative abnormal return (CAR) is:

$$CAR(t_1, t_2) = \sum_{t=t_1}^{t_2} \bar{A}_t \quad (A7)$$

The test statistic TS_j for a j -day event period is derived similarly to that for a single day:

$$TS_j = \frac{\sum_{t=t_1}^{t_2} \bar{A}_t}{\hat{S}(\bar{A}_t) \sqrt{j}} \quad (A8)$$

Appendix B

For each sample firm, we search its corporate website and the global business press for other announcements regarding the firm during the period April 23, 2013 through July 20, 2013. Global business press publications searched are *The Wall Street Journal*, *Herald Sun*, *Globe and Mail*, *Toronto Star*, *Financial Times*, *International Herald Tribune*, *The Daily Telegraph*, *The Times*, and both the US and foreign versions of *Dow Jones News Service* and *PR Newswire*. Announcements that include earnings releases, dividends, changes in the top management team or Board of Directors, and new corporate strategies are identified for each sample firm. For each such announcement, the abnormal return time series is adjusted by excluding the abnormal returns calculated for the affected firm for a three-day period (the day before, day of, and day following the announcement). Table B1 presents a sample of the types of contemporaneous announcements that we use to adjust our abnormal returns. The three-day exclusion period allows for pre- and post-announcement stock market reaction. If the announcement is from print media such as *The Wall Street Journal*, it is possible that information about the announcement was released the day before publication and the stock market reacted the day before publication. If the announcement is released in newswires such as the *Dow Jones News Service* after the stock market has closed, then the market cannot react until the day following the announcement.

Table B1. Examples of the contemporaneous announcements used to adjust the abnormal return time series.

Firm	Date	Event Period	Announcement
N Brown Group	April 24, 2013	1st	Year-end results; dividend raised 5%
VF Corporation	April 26, 2013	1st	1st quarter results; above analyst estimates
Loblaw	May 01, 2013	1st	1st quarter results; dividend hike
Metro AG	May 02, 2013	1st	1st quarter results
Puma	May 14, 2013	2nd	Quarterly results; annual profit warning
Hennes & Mauritz (H&M)	May 15, 2013	2nd	April sales results; up 11%
The Jones Group	July 08, 2013	3rd	Exploring sale of company
Gap Inc.	July 11, 2013	3rd	June sales results; up 8%